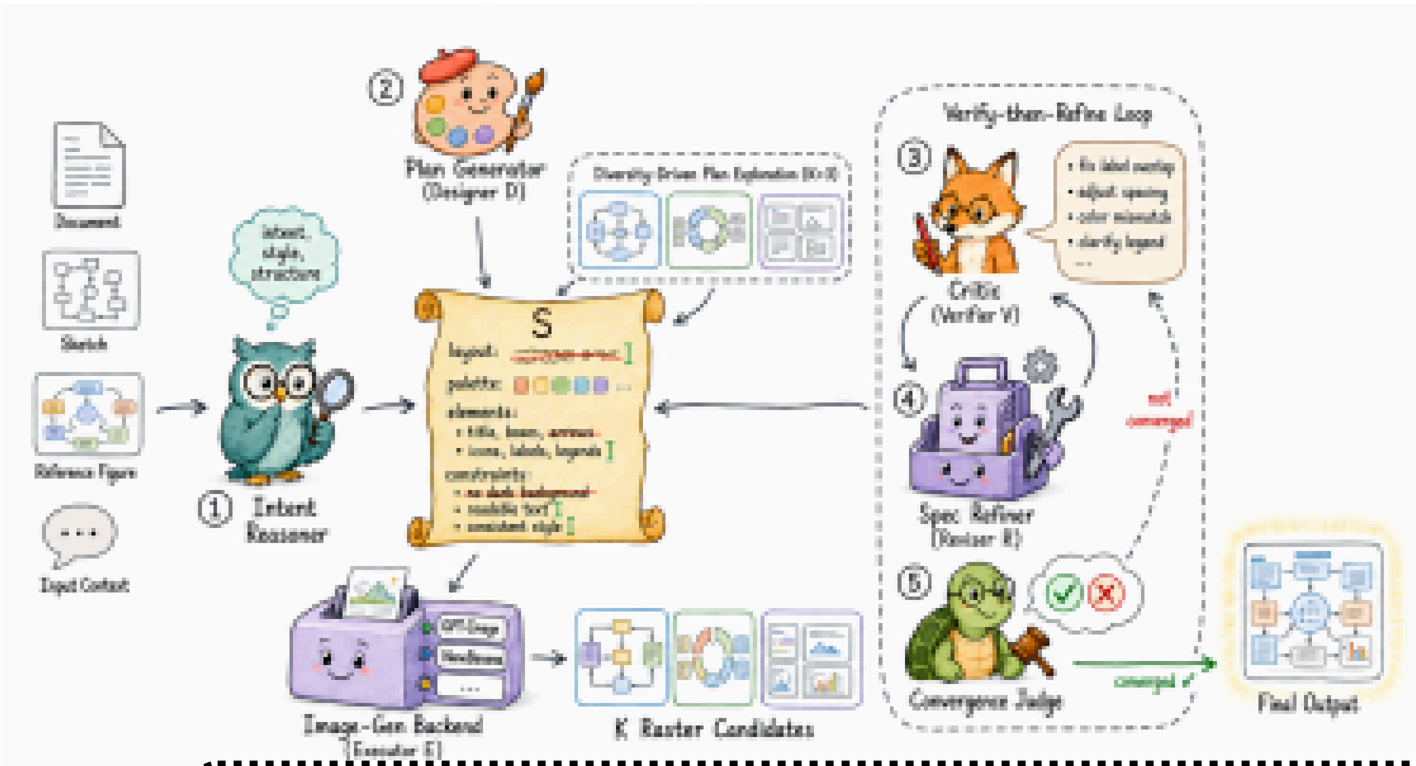
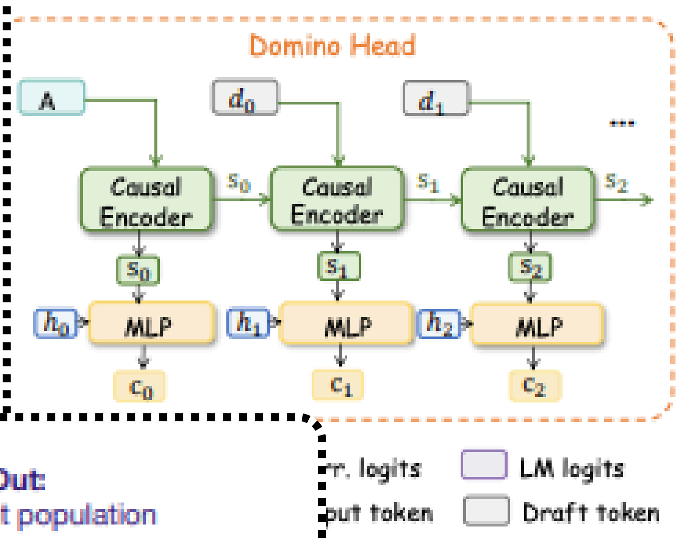
Top 3 Research Papers of the Week





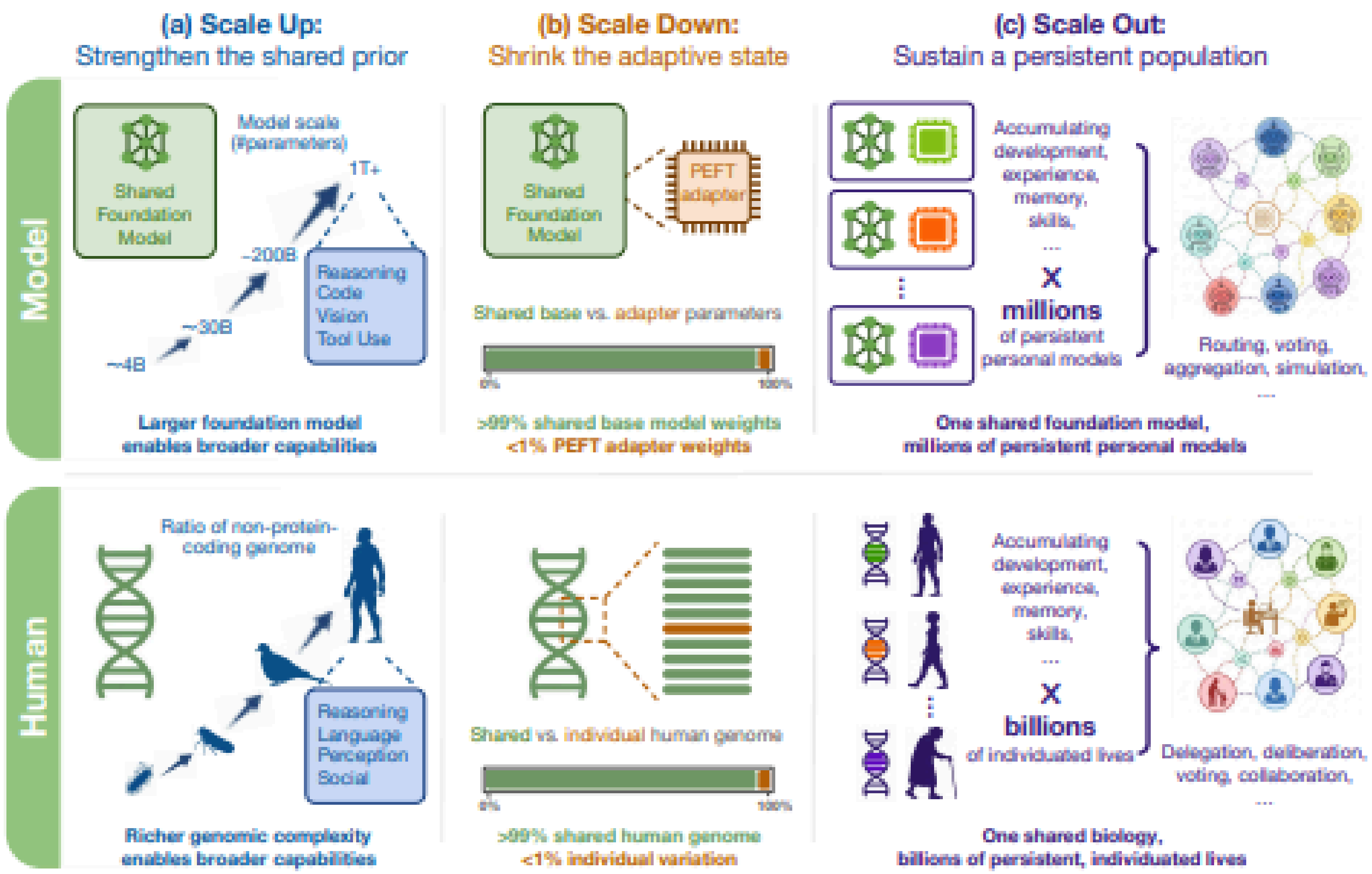
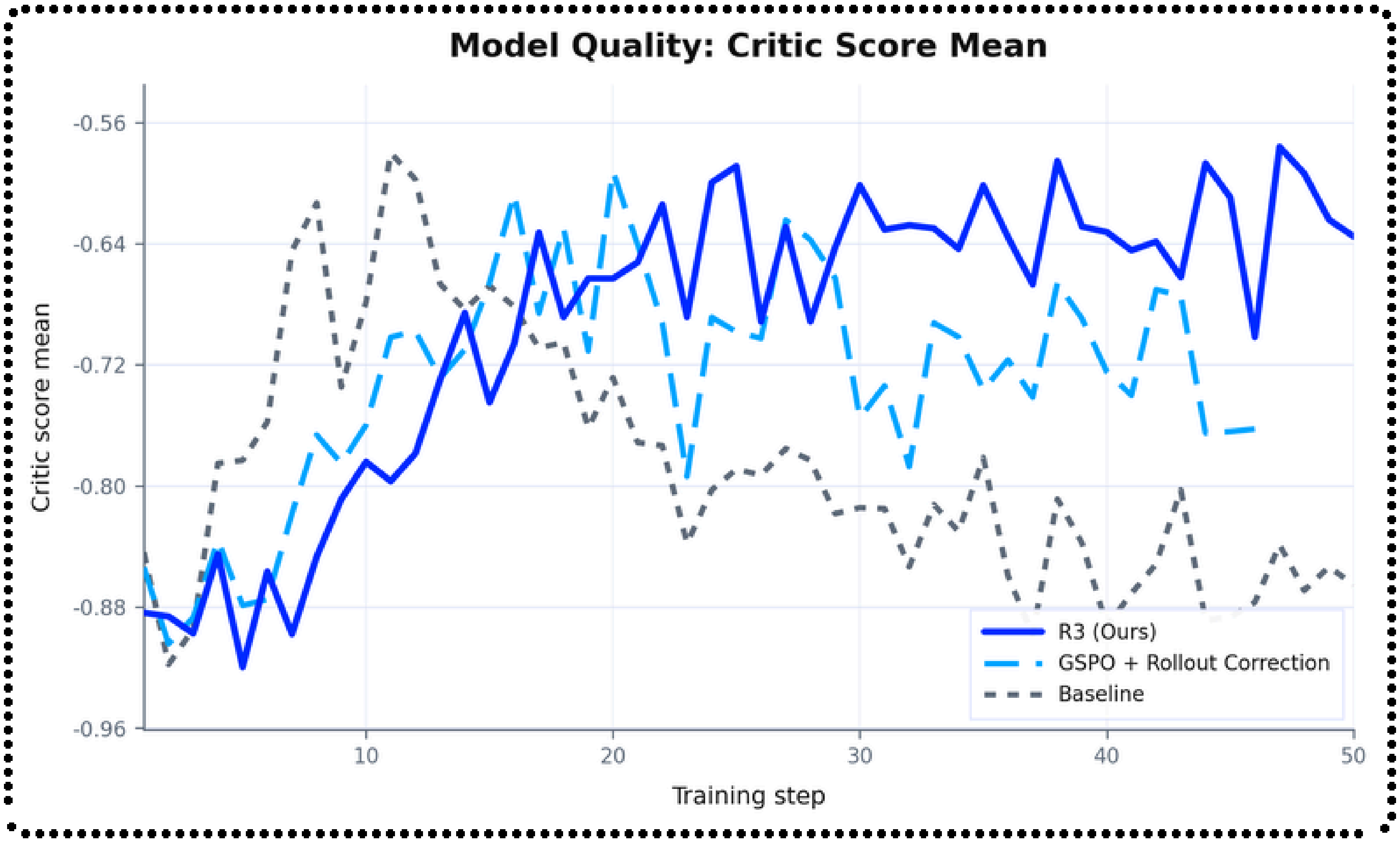


Figure 1 The three scaling axes of PEFT, illustrated through a biological analogy. **(a) Scale Up.** As organisms grow more complex up the tree of life, the fraction of regulatory DNA increases, from roughly 10% in simple organisms to 98% in humans (ENCODE Project Consortium, 2012), enabling richer reasoning, perception, and social behavior. Analogously, larger foundation models unlock capabilities such as reasoning, code, vision, and tool use that make small local updates more powerful. **(b) Scale Down.** Human individuals share more than 99% of their genome (1000 Genomes Project Consortium et al., 2012); the differences that make each person distinct amount to less than 1% of the total. A PEFT adapter occupies a similarly small fraction of the base model weights, and carries the local adaptive state that distinguishes one personal model from another. **(c) Scale Out.** One shared biology supports billions of persistent, individuated human lives, each accumulating its own development, experience, memory, and skills. One shared foundation model can support millions of persistent personal model instances in the same way, each shaped by its own history and each a member of a larger population.

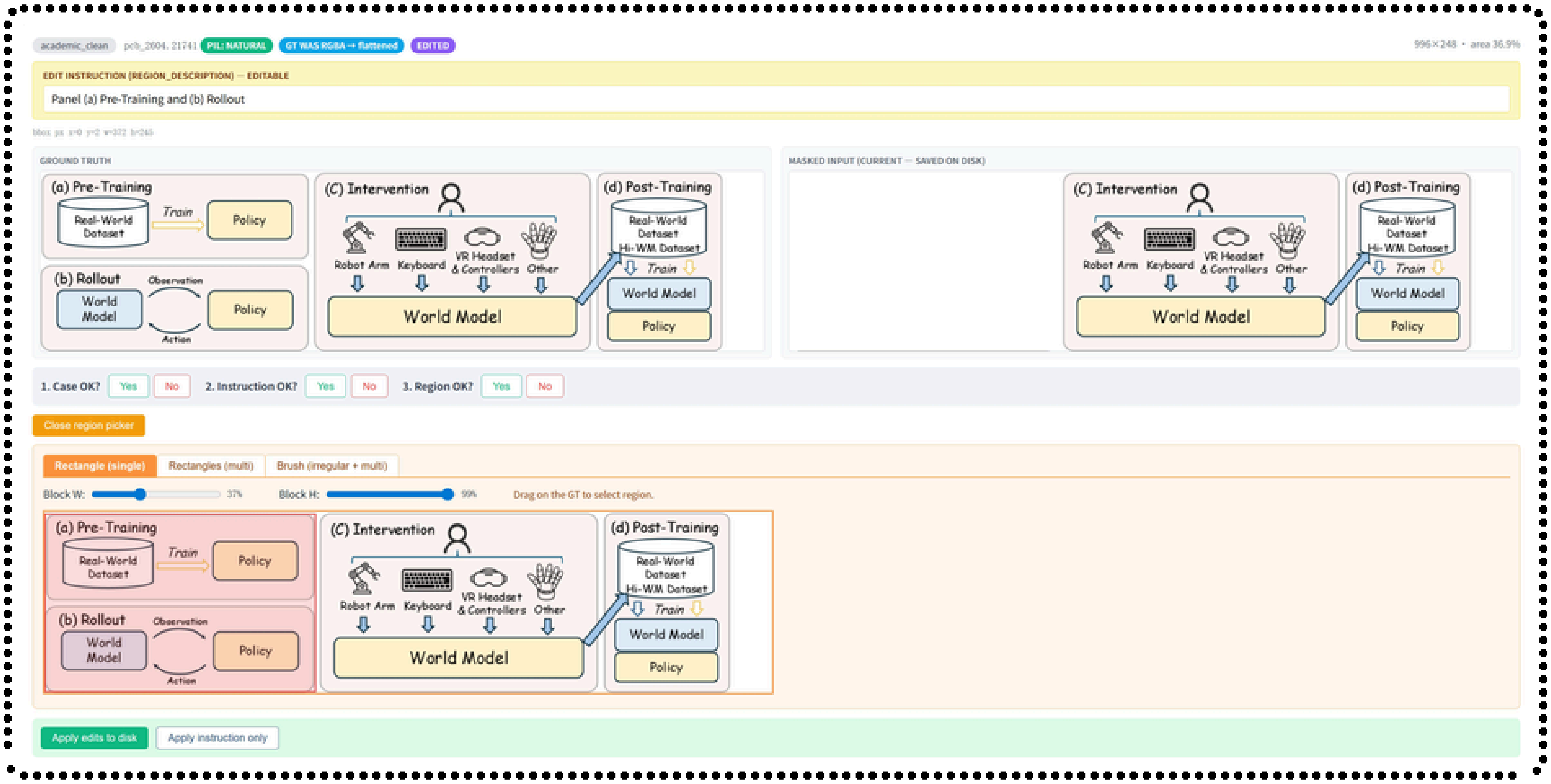
tical way to represent part of that state. The goal is not PEFT itself. The goal is a persistent personal

Scaling PEFT for Personal Models



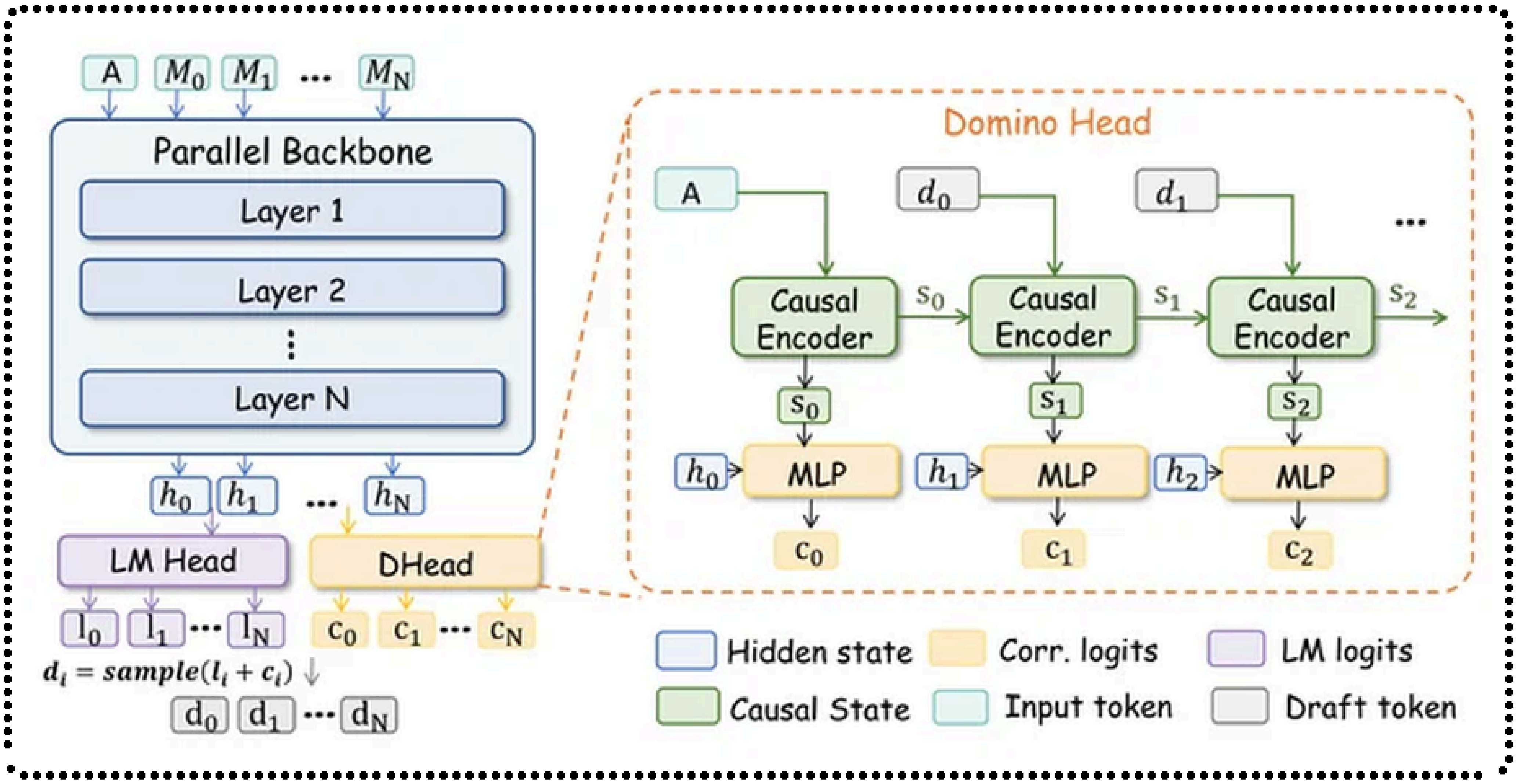
- This paper proposes a three-axis scaling framework for Parameter-Efficient Fine-Tuning (PEFT) to support millions of persistent personal AI models from trillion-parameter shared foundation models, addressing a critical need for scalable personalization.
- The framework explores "Scale Up" (LoRA RL on MoE models), "Scale Down" (minimizing adaptive state with OLORA-tail and δ-mem), and "Scale Out" (managing many instances for collective intelligence).
- LoRA RL reduces GPU budget for trillion-scale models by 90%, and collective intelligence from diverse LoRA variants improves AIME24 accuracy significantly (from 0.3644 to 0.4867 at k=198).

Crafter: Editable Scientific Figure Generation



- CRAFTER introduces a multi-agent harness for generating and editing scientific figures into editable SVGs from diverse inputs, overcoming limitations of existing systems that produce uneditable raster images.
- It uses cooperating agents within a "harness" abstraction for planning, verification, and structured revision, including diversity-driven plan exploration and a verify-then-refine loop for generation and raster-to-SVG conversion.
- CRAFTER significantly outperforms baselines, achieving a 16.61 point higher score on PaperBanana-Bench, and CRAFTEDITOR excels in converting raster figures to editable SVGs with an 8.04 overall score.

Domino: Faster LLM Speculative Decoding



- Domino enhances LLM inference speed by decoupling causal modeling from autoregressive drafting in speculative decoding using a lightweight correction mechanism, balancing draft quality with computational efficiency.
- It combines a parallel draft backbone with a lightweight GRU-based Domino head for prefix-dependent causal correction, utilizing a base-anchored training curriculum and fused Triton kernels for efficient runtime.
- Domino achieves up to 7.92x end-to-end speedup on GSM8K for Qwen3-8B, a substantial improvement over DFlash (5.21x), by increasing average acceptance length by 16.6% with minimal latency.